openalias

Why wheels run backwards on film, and how to recover the true speed.

PAN IIT OSS explainer • higher secondary to advanced undergraduate (levels 4-6)

Why this exists

Anything that looks at a fast process only at fixed instants can be fooled: wheels run backwards on film, and high notes become low ones in a badly recorded sound. This explainer shows the mathematics behind that using sampling, folding and the FFT, so you can build it yourself.

The end goal

Understand the wagon-wheel effect and recover the true frequency. The end goal is to understand the wagon-wheel effect and recover the true frequency from sampled data. The same ideas are used in digital audio, oscilloscopes, vibration measurement and high-speed video.

How the three sessions build up

1. **Session 1: A wheel on camera: slow, frozen or backwards** (Level 1, conceptual interactive model, 60 min).
2. **Session 2: The alias formula and the Nyquist limit** (Level 2, mathematical engineering model, 75 min).
3. **Session 3: Simulate, find the folded peak, recover the true speed** (Level 3, simulation and validation, 90 min).

1 Session 1: A wheel on camera: slow, frozen or backwards

A. Real-world problem

In old westerns the stagecoach wheels seem to turn backwards while the horses gallop forward. Helicopter blades on video sometimes freeze. Nothing is wrong with the wheel or the blades: a camera does not watch continuously, it takes 24 or 30 pictures every second. Between two pictures the wheel can turn many times, and the brain connects the dots in the simplest way. Can we predict exactly what a viewer will see, and which way the wheel seems to go?

B. Concept and theory

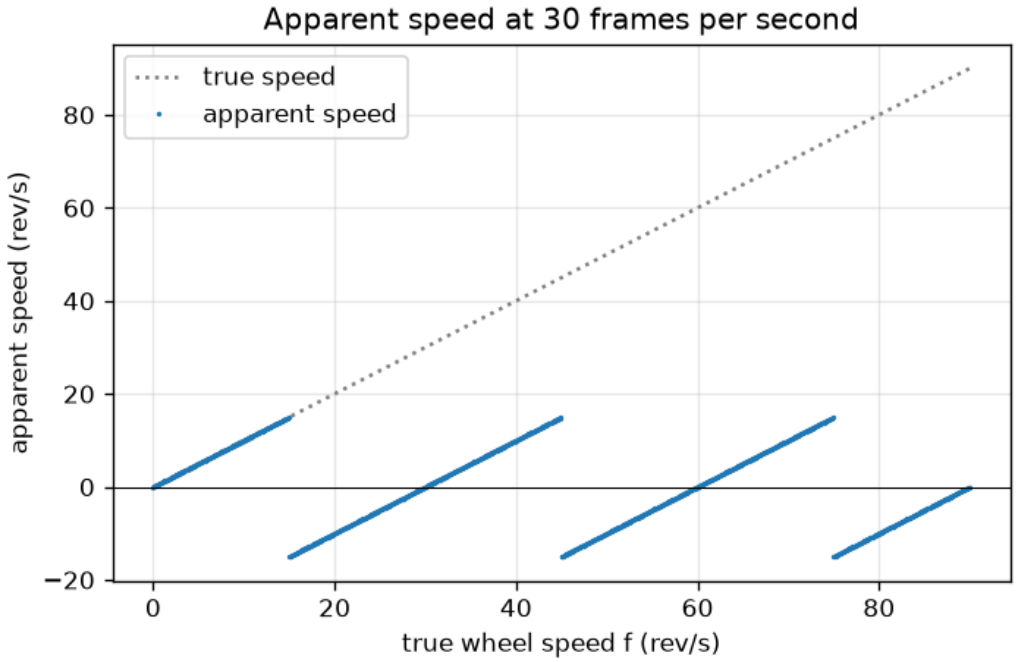
Paint one marker on a wheel that turns f revolutions per second (rev/s). The camera takes fs frames per second, so frames are 1/fs seconds apart. In that time the marker turns step = 360 f / fs degrees, which may be much more than one full turn. A viewer cannot count full turns that happen between frames; they only see where the marker is now and assume the shortest way from the old place to the new one. That shortest signed turn is the step folded into the range -180 to +180 degrees. Multiplying the folded step by the number of frames per second and dividing by 360 gives the apparent speed in rev/s. A negative apparent speed means the wheel seems to turn backwards. Exactly 180 degrees per frame is the flicker limit: both ways look equally short. Adding a whole number of turns to f per frame (that is, adding fs to f) changes nothing the camera can see.

- f: true wheel speed (rev/s)
- fs: frame rate of the camera (frames/s)
- step: true turn between two frames (degrees)
- fold(step): the same turn folded into -180 to +180 degrees (degrees)
- f_app: apparent speed (signed) (rev/s)

$$\text{step} = 360^\circ \frac{f}{f_s}$$

$$\text{fold}(\theta) = \big((\theta + 180^\circ) \bmod 360^\circ\big) - 180^\circ$$

$$f_{\text{app}} = \frac{\text{fold(step)}}{360^\circ} f_s = f - k\, f_s, \qquad k = \text{round}\left(\frac{f}{f_s}\right)$$



Apparent speed against true speed at 30 fps: a sawtooth that repeats every 30 rev/s.

E. Worked example

A wheel turns 28 rev/s and is filmed at 30 frames per second. How far does the marker turn between frames, what does the viewer see, and how fast does the wheel seem to go?

1. Turn between frames: step = 360 x 28 / 30 = 336 degrees.
2. The viewer sees the shortest way: 336 degrees is the same place as 336 - 360 = -24 degrees, so the marker seems to move 24 degrees backwards each frame.
3. Apparent speed = -24 / 360 x 30 = -2 rev/s.
4. Check with f - k fs: k = round(28 / 30) = 1, so 28 - 30 = -2 rev/s. The two methods agree.

Answer. The marker really turns 336 degrees per frame, but is seen to turn 24 degrees backwards: the wheel seems to go backwards at 2 rev/s.

F. Questions

1. **Conceptual.** A wheel turns slightly slower than fs rev/s (for example 29 rev/s at 30 fps). Why does it seem to turn slowly backwards, and not forward?
2. **Numerical.** A wheel turns 33 rev/s and is filmed at 24 frames per second. Find the turn per frame, the apparent turn and the apparent speed. Do a sanity check.
3. **Graph and simulation interpretation.** Using the sawtooth graph for 30 frames per second, for which true speeds between 0 and 60 rev/s does the wheel seem to go backwards, and where does it seem frozen?
4. **Design and optimisation challenge.** A wheel turns 29 rev/s. You may film with 24, 25, 28, 29, 30 or 60 frames per second. Which choice shows it turning slowly forward at no more than 1 rev/s? Which choice shows its true speed? What does the true-speed choice cost?

2 Session 2: The alias formula and the Nyquist limit

A. Real-world problem

A sound card, an oscilloscope and a camera all measure a continuous signal only at fixed instants. Session 1 found the sawtooth by looking at pictures. Engineers need a rule: if the fastest thing in the signal oscillates at f_max hertz, how often must we measure so that no fast signal is mistaken for a slow one? And how do we know which slow signal a fast one will pose as? A table fan with four blades turns 12 rev/s: what frame rate shows it correctly?

B. Concept and theory

Sampling a signal sin(2 pi f t) at the instants t_n = n / fs gives the numbers x[n] = sin(2 pi f n / fs). If we replace f by f + k fs, where k is a whole number, the angle grows by 2 pi k n, a whole number of turns, and the samples do not change. So every frequency in the list ..., f - fs, f, f + fs, f + 2 fs, ... gives the same samples; and because sin(-a) = -sin(a) the negative of each gives the same samples up to a sign. Choose k = round(f / fs). Then f - k fs lies between -fs/2 and +fs/2 and is the smallest of the equivalent frequencies; its size is the alias frequency. It equals f for f below fs/2 and folds back (zigzags) above. The limit fs/2 is the Nyquist frequency. A signal whose highest frequency is f_max is safe only if fs > 2 f_max; exactly 2 f_max is not enough (a sine sampled at its zero crossings gives all zeros). For a wheel with N identical blades or spokes, the picture repeats N times per turn, so the signal frequency is N f, not f; the apparent wheel speed is the signed alias of N f divided by N.

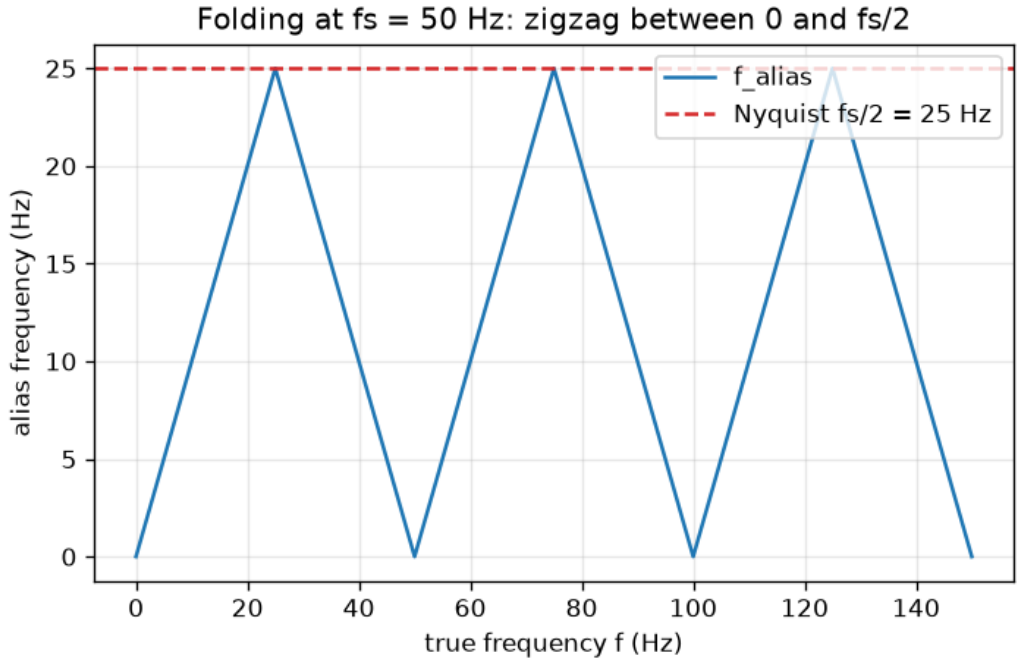
- f: signal frequency (Hz)
- fs: sampling rate (samples/s (Hz))
- k: nearest whole number of sampling rates, round(f / fs) (no unit)
- f_alias: alias frequency, |f - k fs| (Hz)
- f_N: Nyquist frequency, fs / 2 (Hz)
- N: number of identical blades or spokes (no unit)
- f_pattern: frequency of the repeating picture, N f (Hz)

$$x[n] = \sin\left(2\pi f \, \frac{n}{f_s}\right)$$

$$\sin\left(2\pi(f + k f_s)\frac{n}{f_s}\right) = \sin\left(2\pi f \frac{n}{f_s} + 2\pi k n\right) = \sin\left(2\pi f \frac{n}{f_s}\right)$$

$$f_{\text{alias}} = |f - k f_s|\,, \quad k = \text{round}\left(\frac{f}{f_s}\right), \quad 0 \le f_{\text{alias}} \le \frac{f_s}{2}$$

$$f_s > 2f_{\max} \quad \text{(Nyquist condition)}, \qquad f_{\text{pattern}} = Nf$$



Alias frequency against true frequency at fs = 50 Hz, with the Nyquist limit at 25 Hz.

E. Worked example

A 40 Hz sine is sampled at 50 Hz. Find the alias frequency, the signed alias, and check with one sample.

1. Nyquist frequency = 50 / 2 = 25 Hz. Since 40 Hz is above 25 Hz, aliasing will happen.
2. k = round(40 / 50) = round(0.8) = 1.
3. Signed alias = 40 - 1 x 50 = -10 Hz, so the alias frequency is 10 Hz.
4. Check with sample n = 1: x[1] = sin(2 pi x 40 / 50) = sin(2 pi x 0.8) = sin(288 degrees) = -0.951.
5. A -10 Hz sine gives sin(2 pi x (-10) / 50) = sin(-72 degrees) = -0.951. They match.

Answer. The alias frequency is 10 Hz (signed -10 Hz); the 40 Hz samples are exactly those of a -10 Hz sine. The 40 Hz signal poses as a slow 10 Hz one.

F. Questions

1. **Conceptual.** After a signal has been sampled, can any processing of the samples tell a 40 Hz sine from its 10 Hz alias at fs = 50 Hz? Why is the only real cure a change before sampling?
2. **Numerical.** A 130 Hz signal is sampled at 100 Hz. Find the alias frequency and give two other frequencies that give the same samples.
3. **Graph and simulation interpretation.** On the folding graph for fs = 50 Hz, at which true frequencies from 0 to 150 Hz is the alias exactly 10 Hz?

4. **Design and optimisation challenge.** Design: a four-blade fan turns 12 rev/s. A phone can film at 30, 60, 100 or 120 frames per second. Which rates show the true speed and direction? What is the minimum, and what does a higher rate cost?

3 Session 3: Simulate, find the folded peak, recover the true speed

A. Real-world problem

You are handed two video clips of a fan, one filmed at 30 and one at 24 frames per second, and nothing else. In each the marked blade seems to turn backwards, slowly. The fan could be turning at almost any of an endless list of speeds. How do you find the real one, how sure can you be, and what do real cameras (blurry shutters, uneven timing) do to your measurement?

B. Concept and theory

The FFT of N samples taken at rate fs lists the frequencies 0, fs/N, 2 fs/N, ..., fs/2; a pure sine appears as a peak at the alias frequency, so the peak position can be tested against the formula of Session 2 to within one bin, fs/N. A clip of a marker gives an apparent speed with a sign: each frame's marker angle is found, the steps between frames are folded into -180 to +180 degrees and averaged. One clip says f = a1 + k1 fs1 for a whole number k1; a second clip says f = a2 + k2 fs2. Only values that appear in both lists are possible. If both apparent speeds are signed, the pair repeats only after lcm(fs1, fs2) rev/s (whole-number rates), so the answer is unique below that limit; with the same rate twice, or a rate that is a multiple of the other, the limit stays small. If only the sizes |a| are known, we also have to try -a. Two real effects: timing jitter of standard deviation s turns the phase of every sample by a random amount 2 pi f s, which shrinks the coherent peak by exp(-(2 pi f s)^2 / 2); and a shutter open for tau seconds averages the signal over tau, multiplying a sine of frequency f by |sinc(f tau)|, sinc(x) = sin(pi x) / (pi x), a built-in low-pass filter that acts before sampling.

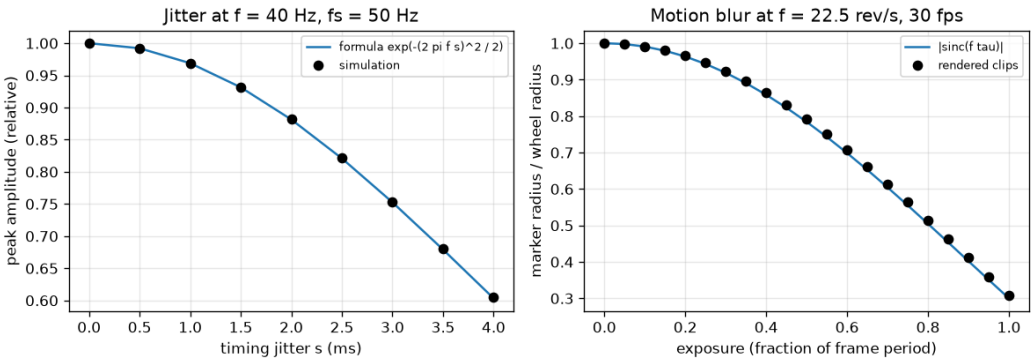
- N: number of samples or frames (no unit)
- fs: sampling or frame rate (Hz)
- a1, a2: signed apparent speeds measured in clip 1 and 2 (rev/s)
- k1, k2: unknown whole numbers (no unit)
- L: unambiguous range, lcm(fs1, fs2) (rev/s)
- s: standard deviation of timing jitter (s)
- tau: exposure time (shutter open) (s)

$$\Delta f_{\text{bin}} = \frac{f_s}{N}$$

$$f = a_1 + k_1 f_{s1} = a_2 + k_2 f_{s2}$$

$$L = \text{lcm}(f_{s1}, f_{s2})$$

$$\frac{A_{\text{measured}}}{A} = e^{-\frac{1}{2}(2\pi f s)^2}, \qquad \frac{A_{\text{exposed}}}{A} = \left|\frac{\sin(\pi f \tau)}{\pi f \tau}\right|$$



Simulation against the jitter formula (left) and the exposure sinc formula (right).

E. Worked example

The 30 fps clip shows the marker at -7.5 rev/s and the 24 fps clip shows it at -1.5 rev/s. The fan turns at less than 60 rev/s. Find its true speed and say how sure you are.

1. Clip 1 allows f = -7.5 + 30 k: ..., -7.5, 22.5, 52.5, 82.5, ...
2. Clip 2 allows f = -1.5 + 24 m: ..., -1.5, 22.5, 46.5, 70.5, ...
3. Values in both lists below 60: only 22.5 (k = 1 and m = 1). Check: -7.5 + 30 = 22.5 and -1.5 + 24 = 22.5.
4. Next common value: 22.5 + lcm(30, 24) = 22.5 + 120 = 142.5, far above the bound, so the answer is unique below 142.5 rev/s.
5. In rpm: 22.5 x 60 = 1350 rpm. Also check the simulation: sampled at 30 Hz the FFT peak is at 7.5 Hz, matching the formula.

Answer. The fan turns 22.5 rev/s (1350 rpm), and this is the only possibility below 142.5 rev/s.

F. Questions

1. **Conceptual.** Why does a long exposure act as an anti-alias filter, and why can this not be repaired after the video is recorded?
2. **Numerical.** A frame clock has 1 millisecond of random timing jitter. By what factor does the peak of a 40 Hz signal shrink? Do the same for a 400 Hz signal. Which signal suffers more and why?
3. **Graph and simulation interpretation.** In the spectrum figure the FFT of the 22.5 Hz signal sampled at 30 Hz peaks at 7.5 Hz. Name four other true frequencies that would give a peak at the same place.
4. **Design and optimisation challenge.** Design: the first camera runs at 30 fps and fan speeds up to 100 rev/s must be recovered uniquely from signed measurements. Which second frame rate would you choose from 15, 20, 24, 25 and 60? Justify the choice.

4 Real-world mini-project

How fast does the fan really turn?. A fan with one blade marked in red was filmed twice: once at 30 frames per second and once at 24 frames per second. At 30 fps the marker seems to turn backwards, at 24 fps it also seems to turn backwards but much more slowly. The clips are the function `mystery_fan(fs)` of Session 3 (do not read the true speed from its code). The fan is known to turn at less than 60 revolutions per second. Measure both clips, recover the true speed, and confirm it with a third clip filmed at 25 fps.

5 The path ahead

1. **Session 4: Noise-proof recovery.** A recovery function that takes noisy apparent speeds and returns the most likely true speed with an uncertainty.
2. **Session 5: A third clip.** A recovery function for three clips with a unique answer over a wide range.
3. **Session 6: Fans with many identical blades.** A clip renderer and a measurement function for N identical blades, with one blade tinted to tell them apart.
4. **Session 7: A fan that spins up.** A tracker that follows the folded apparent speed through time and reconstructs the true speed curve.
5. **Session 8: Real cameras: variable frame rate and rolling shutter.** A simulator with frame timestamps from a clock with 0.5 percent jitter and a line-by-line shutter, plus a correction.

6 Answers and hints

Session 1

1. **Conceptual.** Hint: Between two frames the marker turns almost one full turn. Compare it with a full turn. Solution: Between two frames the marker turns 348 degrees, which is 12 degrees short of a full turn. The eye ignores the full turn and sees the marker land 12 degrees before where it started, so it seems to move 12 degrees backwards per frame: -12 / 360 x 30 = -1 rev/s. A wheel slightly faster than fs would land slightly past its start and seem to move slowly forward.
2. **Numerical.** Hint: step = 360 f / fs, then fold into -180 to +180 degrees. Solution: step = 360 x 33 / 24 = 495 degrees. Folded: 495 - 360 = 135 degrees. Apparent speed = 135 / 360 x 24 = +9 rev/s (forward). Check: f - k fs with k = round(33 / 24) = round(1.375) = 1 gives 33 - 24 = 9 rev/s. Sanity: the apparent speed can never exceed fs / 2 = 12 rev/s in size, and 9 is below it.
3. **Graph and simulation interpretation.** Hint: Backwards means the curve is below zero. Solution: The curve is below zero for $15 < f < 30$ rev/s and for $45 < f < 60$ rev/s: backwards. It touches zero at f = 0, 30 and 60 rev/s: frozen. At exactly f = 15 and 45 rev/s the apparent speed is 15 in size, the flicker limit, and the direction cannot be told. Units: rev/s on both axes.
4. **Design and optimisation challenge.** Hint: Compute f - fs x round(f / fs) for each frame rate. Solution: The apparent speeds are +5 (24 fps), +4 (25 fps), +1 (28 fps), 0 (29 fps, frozen), -1 (30 fps, backwards) and +29 (60 fps). Only 28 fps shows slow forward motion of at most 1 rev/s. Only 60 fps shows the truth, because 60 is above twice 29 (Session 2 explains this rule). Trade-off: doubling the frame rate doubles the amount of data and needs more light per frame.

Session 2

1. **Conceptual.** Hint: Look at the sample values you derived for the worked example. Solution: No. The samples of 40 Hz and -10 Hz are identical numbers, so no function of the samples can tell them apart. The information is lost at the instant of sampling. The cure must act before: sample faster (fs above 2 f_max) or remove frequencies above fs/2 with a low-pass (anti-alias) filter first.
2. **Numerical.** Hint: k = round(130 / 100); then use f -/+ fs and the sign symmetry. Solution: k = round(1.3) = 1, so f_alias = |130 - 100| = 30 Hz (signed +30). Frequencies with the same samples: 30 + 100 = 130, 30 - 100 = -70 (so 70 Hz up to a sign), 30 + 200 = 230, and

the negatives. So 70 Hz and 170 Hz alias to 30 Hz as well. Sanity: 30 is between 0 and fs/2 = 50.

3. **Graph and simulation interpretation.** Hint: Draw a horizontal line at 10 Hz and read where it crosses the zigzag. Solution: The zigzag crosses 10 Hz at f = 10, 40, 60, 90, 110 and 140 Hz. They come in pairs symmetric about each multiple of 50: 50 -/+ 10, 100 -/+ 10, 150 - 10 (and 0 + 10).
4. **Design and optimisation challenge.** Hint: The picture repeats N f = 48 times per second. Apply the Nyquist condition to the pattern. Solution: The pattern frequency is 4 x 12 = 48 Hz, so fs must be above 2 x 48 = 96 frames per second: only 100 and 120 fps work (apparent +12 rev/s). At 30 fps the fan seems to turn at -3 rev/s (48 - 60 = -12 Hz, divided by 4) and at 60 fps also -3 rev/s. Exactly 96 would sit on the limit. A higher rate costs more data, more light per frame (shorter exposure) and storage; 100 fps is the cheapest choice with a margin of 4 percent only, 120 fps gives a safe margin of 25 percent.

Session 3

1. **Conceptual.** Hint: Averaging a fast oscillation over a time much longer than one cycle gives about zero. Solution: During the exposure the sensor adds up the light, which averages the signal over tau seconds; a sine of frequency f is multiplied by |sinc(f tau)|, which is small once f tau is above 1. So fast motion is smeared away before the camera samples it and cannot fold. Afterwards the blur has already removed the information (a fully blurred marker has no position left), so no processing can bring it back.
2. **Numerical.** Hint: factor = exp(-(2 pi f s)^2 / 2). Solution: At 40 Hz: 2 pi x 40 x 0.001 = 0.2513, squared 0.0632, half 0.0316, factor exp(-0.0316) = 0.969, a loss of 3.1 percent. At 400 Hz: 2.513, squared 6.32, half 3.16, factor exp(-3.16) = 0.042, the peak is almost gone. The higher true frequency suffers more because the same timing error is a larger fraction of one cycle. It is the true frequency, not the alias, that counts.
3. **Graph and simulation interpretation.** Hint: The unsigned alias 7.5 Hz comes from f = 7.5 + 30 k and f = -7.5 + 30 k. Solution: 7.5 + 30 = 37.5, 22.5 (given), 52.5, 67.5, 82.5 and 7.5 itself. So the peak position alone cannot tell which one it was; a second frame rate is needed. The bin spacing here is 30 / 300 = 0.1 Hz, so the peak is read to about 0.05 Hz.
4. **Design and optimisation challenge.** Hint: The unique range is lcm(30, fs2); it must exceed 100. Solution: lcm(30, 15) = 30, lcm(30, 20) = 60 and lcm(30, 60) = 60 all fail (they are too small because the rates share large factors). lcm(30, 24) = 120 and lcm(30, 25) = 150 work. Choose 25 fps: the largest range (150 rev/s) gives margin if the true speed is a little above 100. Trade-off: the ladder rungs are spaced 30 and 25 rev/s apart, and the common rungs near each other differ by a few rev/s only when the rates are close, so measurement noise must stay below that difference; very close rates (such as 30 and 29) give a huge range but need very accurate measurements.

7 How to join

Read `CONTRIBUTING.md` in the repository, then pick an issue labelled `good first issue`. Repository: https://github.com/KrunalPanchal1995/openalias